Spring 2026 Practice Midterm Exam

Foundations of Data Science

Name

Instructions

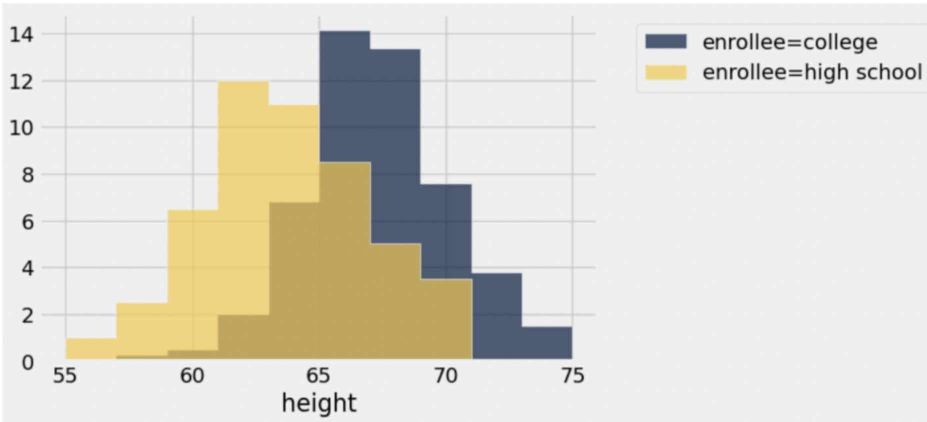
- Open the Midterm Exam Reference Guide (linked in Canvas). We will provide you with a new copy of this reference to use on the exam.
- Previews of most of the tables you will work with in this exam are at the end of exam in the Table Reference section.
- You can assume the following code has been run, when you are writing your Python code:

```
from datascience import *
import numpy as np
import matplotlib+
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

- Select the correct response(s) or provide a written response depending on the question type. If a prompt implies you should write code, then you can provide your own code or use the provided template. Try to provide your responses in the template blanks or boxed spaces provided. If you find that you need additional space, write your extended response(s) on a blank sheet of paper and number them, so we can connect your response to the question.
- Once you have made an attempt on the exam, upload a PDF of your attempt to Canvas for a Complete score. Sample solutions will be released before the Midterm Exam Study Session.
- You will be asked a few questions about Project 1. Some of the questions may be about the code you wrote, and some of the questions may have you interpret the results and visualizations.

This practice exam contains questions from past UC Berkeley DATA 8 midterm exams in addition to a few additional or modified questions.

1. **Heights** Dylan, a Data 8 student, wonders if there is a difference in heights between his high school and college classmates. Among Data 8 students, he collects a random sample of 100 high school students and a random sample of 200 college students, and organizes his data in a table called `heights`. He created the following histogram based on the data in `heights` where the x-axis of the histogram has a range of $[55, 75]$ and each bin is 2 inches in width.



(a) Unfortunately, Dylan forgot to include the y-axis label in his histogram. Which of the following is an appropriate y-axis label? Select one.

○ Percentage of students

○ Percentage of students per height

√ **Percentage of students per inch**

○ Rate of change in height

(b) Which of the following is closest to the total number of individuals in our table that are greater than or equal to 61 inches, but less than 63 inches in height? Select one.

○ 14

○ 24

○ 28

√ **32**

(c) Which of the following conclusions can you draw from Dylan's overlaid histograms? Select all that apply.

☐ Assuming we know the height of each bin, we can calculate the percentage of individuals in any range of heights.

√ **College Data 8 students are generally taller than high school Data 8 students.**

☐ The area of both histograms together sum to 100%, so the area of the blue and yellow histograms are $\frac{100}{300} \times 100$ and $\frac{200}{300} \times 100$, respectively.

☐ There is an association between a Berkeley student's height and age.

☐ College students enrolled in Data 8 are generally taller than high school students in Data 8 because they are older.

2. The empirical distribution of a categorical variable contains the unique values of the variable along with how often each value is observed.

√ **True**

○ False

3. **Real Estate.** A real estate company has a data set of all of their buildings, with four attributes for each building: its size (in square feet), its type (residential or commercial), its ZIP code, and its estimated value if sold (sale price in dollars).

   (a) The standard visualization to understand the distribution of building types is,

   √ **A bar chart** ◯ A line plot ◯ A scatter plot ◯ Overlaid histograms

   (b) The standard visualization to check for an association between building size and building value is,

   ◯ A bar chart ◯ A line plot √ **A scatter plot** ◯ Overlaid histograms

   (c) The standard visualization to compare the distributions of the estimated values (sale price in dollars) of the two types of buildings is,

   ◯ A bar chart ◯ A line plot ◯ A scatter plot √ **Overlaid histograms**

   (d) Select all that are correct:

   ☐ The size attribute is a categorical variable.

   √ **The type attribute is a categorical variable.**

   ☐ The ZIP code attribute is a numerical variable.

   √ **The estimated value attribute is a numerical variable.**

4. **Hip Pain.** In an effort to investigate whether a new treatment is effective at reducing hip pain, researchers randomly sampled 100 patients from a medical group that used that new treatment, as well as other older treatments. They asked each patient whether or not they had received the new treatment or an older treatment, as well as whether or not they had experienced a reduction in pain. Overall, the results showed that those who received the new treatment observed a significant reduction in pain. Which of the following can be concluded from this passage? Select all that apply.

   √ **This is an observational study.**

   ☐ This an experiment.

   ☐ This is a randomized controlled experiment.

   ☐ The new treatment causes a reduction in pain.

   √ **There is a significant association between the new treatment and pain reduction.**

5. **LinkedIn.** According to a recent survey, 28% of surveyed adults in the United States use LinkedIn. For the sake of this question, assume that the chance of a randomly sampled adult in the United States being a LinkedIn user is 28% (independently of all others).

   (a) For which sample size below is there a higher chance that a random sample of that size will contain a percent of LinkedIn users of more than 50%?

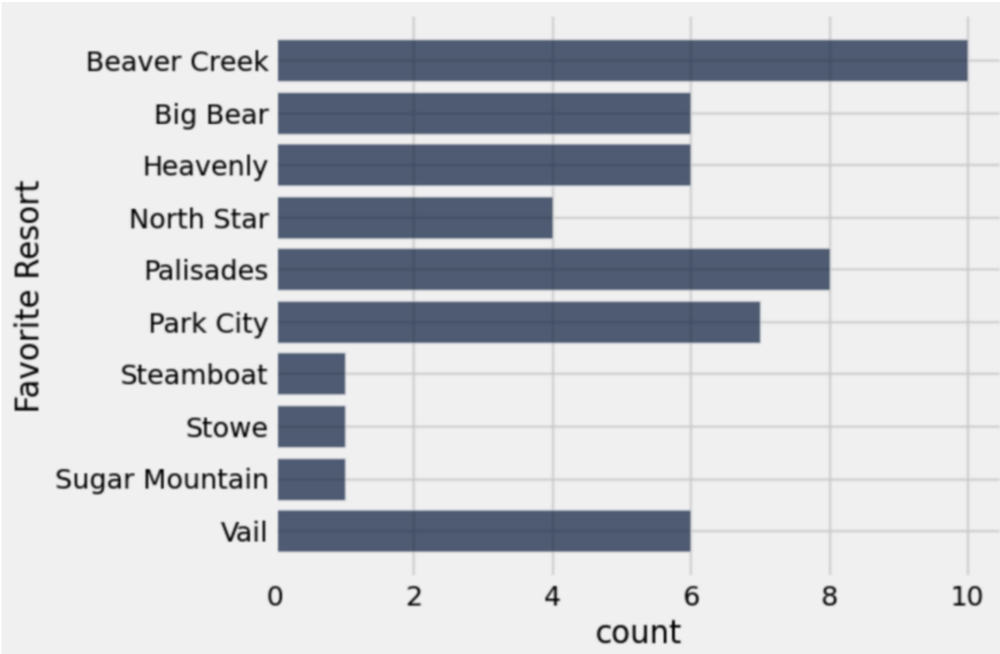
   √ **20** ◯ 1,000

   (b) According to the Law of Large Numbers (Law of Averages), with a smaller sample size the percentage of surveyed adults in that sample that use LinkedIn is more likely to be closer to 28% than a larger sample size.

   ◯ True √ **False**

6. **Skiers** The table `skiers` previewed on the Table Reference page contains information about the preferences of several skiers from a convenience sample of Data 8 staff.

(a) Fill in the blanks to generate the following bar chart showing the popularity of everyone's Favorite Resort given in the skiers table. You may find the axis labels helpful.




```
favorite_counts = skiers.________________(____________________)
__________________.__________________(_______________________________)
```

Sample Solution:

```
favorite_counts = skiers.group("Favorite Resort")
favorite_counts.barh("Favorite Resort")
```

(b) The height of the skiers was accidentally inputted into the `skiers` table as strings. Create a function called `str_to_int` that takes an input of a skier height in the format of a string and outputs the height as an integer. For example `str_to_int('71')` will output `71`.

Sample Solution:

```
def str_to_int(height):
    return int(height)
```

(c) Update the table `skiers` so that the column `'Height (in)'` has integer values, not string values.

Hint. Consider using the `apply` and `with_column` table methods.

Sample Solution:

```
heights = skiers.apply(str_to_int, 'Height (in)')
skiers = skiers.with_column('Height (in)', heights)
```

(d) Compute the average skier height for all the skiers in the `skiers` table.

Sample Solution:

```
np.average(skiers.column('Height (in)'))
```

7. If `customers` is a Table with a column called `'Age'`, why does the following code produce an error?

```
customers.where('Age', are.above(24)) / customers.num_rows
```

Sample Solution: This produces an error because `customers.where('Age', are.above(24))` prodcues another Table and `customers.num_rows` produces an integer, yet the division of a Table and an integer is not defined.

8. **Food Orders** The table `orders` previewed on the Table Reference page contains information about food orders that members of Data 8 Course Staff have made this semester.

(a) Create a scatter plot to visualize the relationship between how much the order costs versus how the user rated the order.

```
orders.__________________(_________________, __________________)
```

Sample Solution:

```
orders.scatter("Total", "Rating")
```

(b) Assign the variable `usually_friends` to `True` if ordering with friends is more common than not ordering with friends and `False` otherwise.

```
with_friends = ________________, True).num_rows
without_friends = __________________.__________________ - ___________________
usually_friends = with_friends ___________________ without_friends
```

Sample Solution:

```
with_friends = orders.where("With Friends", True).num_rows
without_friends = orders.num_rows - with_friends
usually_friends = with_friends > without_friends
```

(c) Assign the variable `frugal_user` to the name of the user who has spent the least over the entire semester:

```
frugal_user = (
    orders.group(__________________, __________________)
    .sort(___________________, descending=___________________)
    .column("User").item(0)
)
```

Sample Solution:

```
frugal_user = (
    orders.group("User", sum)
    .sort("Total sum", descending=False)
    .column("User").item(0)
)
```

9. **Smoothies.** Amy makes a smoothie every morning before going to school. To make her smoothie, she goes into the refrigerator and picks two pieces of fruit completely at random (first one, then the other). Suppose there are 2 pears, 5 mangos, 4 kiwis, and 1 peach in her refrigerator each morning when she goes to make her smoothie, and that each fruit is equally likely to be chosen.

For each part below, write an expression that evaluates to the probability described. You do not need to simplify any arithmetic.

(a) The probability that her next smoothie contains only kiwis.

Sample Solution: `(4 / 12) * (3 / 11)`

(b) The probability that her next smoothie has at least one mango.

> **Sample Solution:** `1 - (7 / 12) * (6 / 11)`

(c) The probability that her next smoothie has a kiwi and a pear.

> **Sample Solution:** `(4 / 12) * (2 / 11) + (2/ 12) * (4 /11)`

(d) The probability that her next smoothie contains only peaches.

> **Sample Solution:** `(1 / 12) * (0 / 11)` or `0`

10. **Count Occurrences.** Which of the following functions correctly returns the number of occurrences of a specific value in a given array? For example,

```
count_arr_occurences(make_array(0,1,0,5,1), 1)
```

should evaluate to 2 and

```
count_arr_occurences(make_array("a", "b", "c"), "c")
```

should evaluate to `1`. Select all that apply.

☐
```
def count_arr_occurences(arr, value):
    count = 0
    for i in np.arange(value):
        if arr.item__ == value:
            count = count + 1
    return count
```

√
```
def count_arr_occurences(arr, value):
    count = 0
    for x in arr:
        if x == value:
            count = count + 1
    return count
```

☐
```
def count_arr_occurences(arr, value):
    return arr == value
```

√
```
def count_arr_occurences(arr, value):
    return np.sum(arr == value)
```

11. **Chocolates and Nutrition.** The two tables called `chocolates` and `nutrition` show some inventory data about a certain store's chocolates and the caloric data for each type of chocolate.

Here is a list of table method calls referencing these tables:

A. `chocolates.group("Shape")`
B. `chocolates.group("Shape", max)`
C. `chocolates.group(["Shape", "Color"], max)`
D. `chocolates.pivot("Color", "Shape", "Unit Price ($)", max)`
E. `chocolates.join("Color", nutrition, "Type")`
F. `chocolates.group(["Shape", "Color"])`

Write the letter of the table method call in the row that matches the description of the resulting table.

Table Method Letter	Columns in Table	# of Rows
B	`Shape`, `Color max`, `Inventory max`, `Unit Price ($) max`	2
D	`Shape`, `Dark`, `Milk`, `White`	2
C	`Shape`, `Color`, `Inventory max`, `Unit Price ($) max`	4
E	`Color`, `Shape`, `Inventory`, `Unit Price ($) max`, `Calories`	6
A	`Shape`, `count`	2
F	`Shape`, `Color`, `count`	4

12. Complete the code below to simulate rolling a fair six-sided die 10,000 times and estimate the probability of rolling a 4 or higher.

```
count = 0
num_reps = 10_000
die = np.arange(1, 7)

for i in np.arange(____________________):
    random_die_roll = np.random.choice(____________________)
    if _______________________:
        count = count + 1

prob_4_or_higher = ________________________
```

Sample Solution:

```
count = 0
num_reps = 100
die = np.arange(1, 7)

for i in np.range(num_reps):
    random_die_roll = np.random.choice(die)
    if random_die_roll >= 4:
        count = count + 1

prob_4_or_higher = count / num_reps
```

13. **Wordle.** In the game of Wordle, a player guesses up to 6 words until they either correctly guess the secret word of the day or run out of guesses. Their guess count is either the number of guesses needed to guess the correct word (1 through 6) or X if all 6 guesses were incorrect. One thousand CCSF students played Wordle yesterday and the proportion of these students with each guess count is in the table below. We also created an array of these proportions called `students`.

1	2	3	4	5	6	X
0.0	0.17	0.33	0.27	0.20	0.02	0.01

```
students = make_array(0.0, 0.17, 0.33, 0.27, 0.20, 0.02, 0.01)
```

Wordle's creator, Josh Wardle, sent us the proportion of guess counts for all players who tried to guess yesterday's word and the table of proportions is below. The array called `everyone` contains all of these proportions.

1	2	3	4	5	6	X
0.0	0.09	0.25	0.32	0.28	0.03	0.03

```
everyone = make_array(0.0, 0.09, 0.25, 0.32, 0.28, 0.03, 0.03)
```

(a) What best describes the table for the students? Choose one.

- ○ Probability Distribution
- √ **Empirical Distribution**

(b) What is one way to simulate randomly selecting 1,000 individuals from the population of individuals that played Wordle yesterday? Choose one.

- ○ `sample_proportions(1000, students)`
- √ `sample_proportions(1000, everyone)`
- ○ `sample_proportions(1000, make_array('1', '2', '3', '4', '5', '6', 'X'))`
- ○ `sample_proportions(1000, make_array(1/7, 1/7, 1/7, 1/7, 1/7, 1/7, 1/7))`

(c) You would like to test whether the distribution of guess counts among the 1,000 students who played Wordle is different from the distribution provided by Josh Wardle. What test statistic could be used to run this hypothesis test?

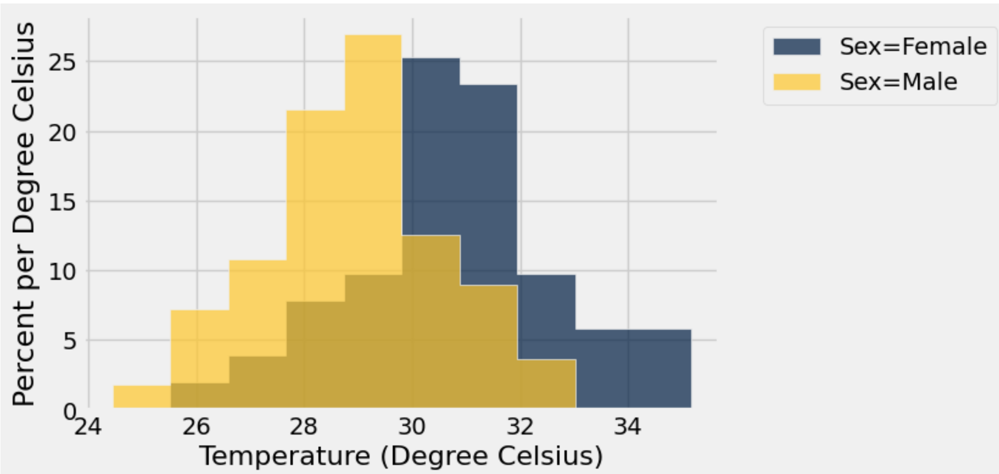
Sample Solution: The TVD can be used for this test.

(d) If you assume the distribution provided by Josh Wardle is similar for tomorrow, what is the chance that a randomly selected Wordle player will guess the word in less than 4 guesses?

Sample Solution: `0.00 + 0.09 + 0.25`

14. **Turtles.** Incubation temperature is an important factor in successful hatching of a baby turtle from an egg. Ellen read that the temperature at which an egg is incubated influences whether the turtle will be male or female. She loves turtles and is wondering whether this claim is true, or whether any observed differences might just be due to chance. She collects data on a random sample of 100 turtles and records both the incubation temperature (in Celsius) and the sex of the turtle that hatches in the table `turtles`.

Ellen decides to visualize her data before doing any inference. She creates the following overlaid histograms showing the distributions of the incubation temperatures for the female and male turtles.



Ellen would like to perform a test to see whether females in the population in general have higher incubation temperatures than males, or if the observed difference in distributions is due to chance.

(a) Which of the following is a valid null hypothesis for this test?

- ◯ Females in the sample in general have higher incubation temperatures than males.
- ◯ There is no difference in incubation temperature between male and female turtles in the sample. Any difference we see in the population is due to chance.
- ◯ Females in the population in general have higher higher incubation temperatures than males.
- ◯ Males in the population in general have higher incubation temperatures than females.
- √ **There is no difference in incubation temperature between male and female turtles in the population. Any difference we see in the sample is due to chance.**

(b) Which of the following is a valid alternative hypothesis for this test?

- ◯ Females in the sample in general have higher incubation temperatures than males.
- ◯ There is no difference in incubation temperature between male and female turtles in the sample. Any difference we see in the population is due to chance.
- √ **Females in the population in general have higher higher incubation temperatures than males.**
- ◯ Males in the population in general have higher incubation temperatures than females.
- ◯ There is no difference in incubation temperature between male and female turtles in the population. Any difference we see in the sample is due to chance.

(c) Ellen uses the difference between average incubation temperatures, defined by "female average minus male average" as her test statistic. In the following code cell, write a definition for the function `diff_of_means` that will take a table and a group label as arguments and returns the value of the test statistic. The table used as input (like `turtles`) has two columns – one of labels and one of numerical values.

```
def diff_of_means(tbl, group_col):
   means_table = __________________________
   means_array = __________________________
   return _________________________
```

Sample Solution:

```
def diff_of_means(tbl, group_col):
    means_table = tbl.group(group_col, np.mean)
    means_array = means_table.column(1)
    return means_array.item(0) - means_array.item(1)
```

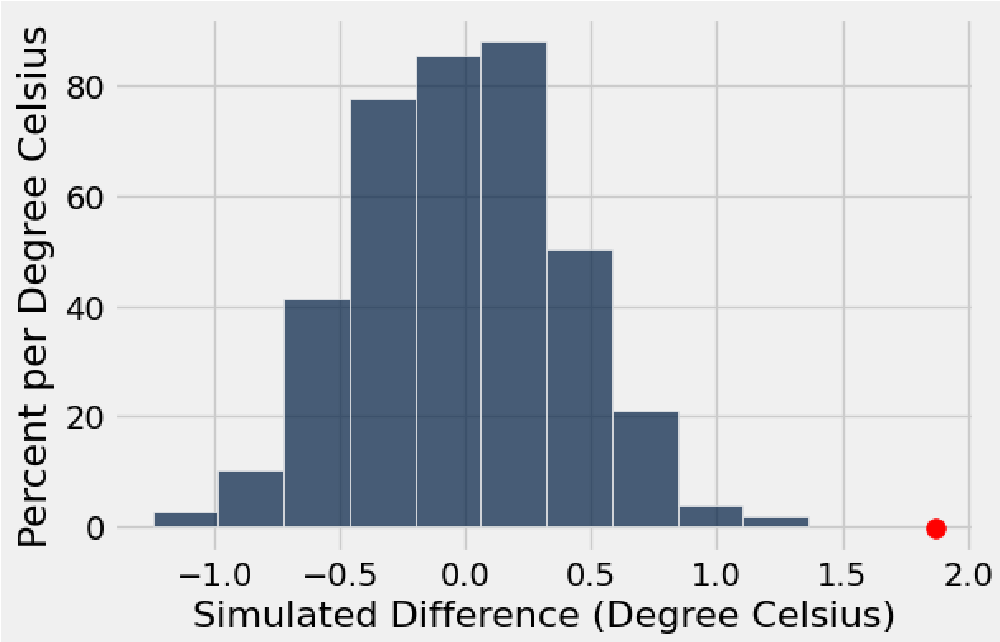
(d) Next, she simulates generating data many times under the null hypothesis. Complete the code below which will run a permutation of the data and calculate the test statistic 1,000 times, and store the results in an array called `diffs_array`.

```
diffs_array = _______________________
repetitions = 1000
for ______________________________:
    resampled_table = turtles.______________________
    shuffled_labels = resampled_table.column('Sex')
    shuffled_table = turtles.with_column('Sex', shuffled_labels)
    shuffled_diff = __________________(shuffled_table, 'Sex')
    diffs_array = _________________________
```

Sample Solution:

```
diffs_array = make_array()
repetitions = 1000
for i in np.arange(repetitions):
    resampled_table = turtles.sample(with_replacement = False)
    shuffled_labels = resampled_table.column('Sex')
    shuffled_table = turtles.with_column('Sex', shuffled_labels)
    shuffled_diff = diff_of_means(shuffled_table, 'Sex')
    diffs_array = np.append(diffs_array, shuffled_diff)
```

(e) Ellen ran `diff_of_means(turtles, 'Sex')` to get an observed value for the test statistic and assigned it to `observed_statistic`. The histogram below shows the distribution of the 1,000 simulated differences along with the observed difference, plotted with a dot.



Using `diffs_array` and `observed_statistic`, write code that will calculate the empirical p-value based on the simulation.

Sample Solution:

```
np.mean(diffs_array >= observed_statistic)
```

(f) Based off of the visualization, what can you conclude about Ellen's test? Assume a p-value cutoff of 5%.

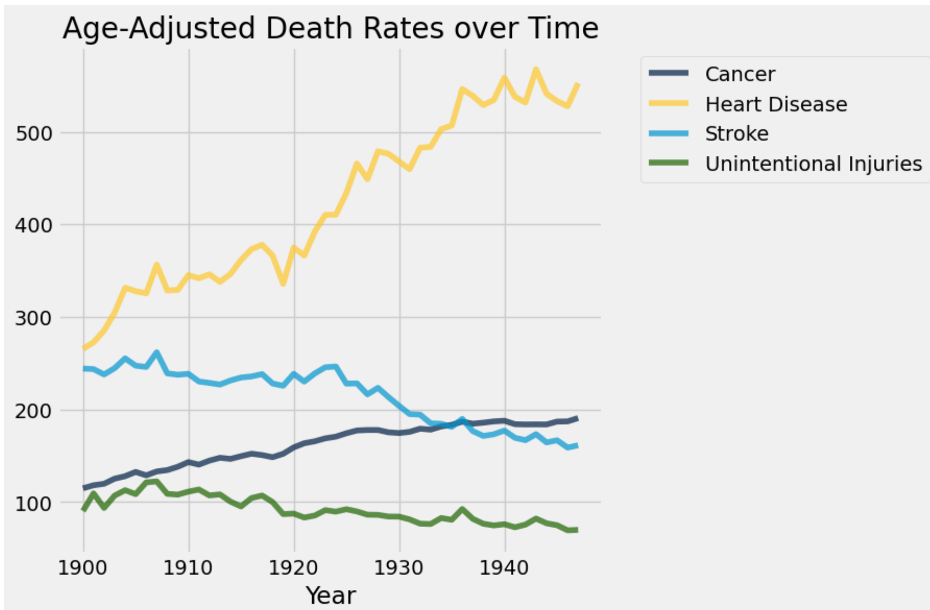
- ◯ The data are consistent with the null hypothesis.
- √ **The data are consistent with the alternative hypothesis.**
- ◯ There is not enough evidence to make a conclusion.

15. Suppose a hypothesis test is proposed and we already know that the null hypothesis is true. If 500 researchers each independently collects a sample of the same size to carry out an experiment and they all use 1% as their p-value cutoff, we should expect around 5 of them to reject the null.

- √ **True**
- ◯ False

16. **Midterm Project.** You have been working on the Midterm Project where you examined three health studies that were conducted over the last century. There was data about the leading causes of death over the last 120+ years, a study about how cholesterol levels correlate with heart disease, and the National Diet-Heart Study. In this section, you will answer several questions that come directly from that project and the material covered in it.

(a) Given the table `causes_for_plotting` table, write code that will produce the following graphic showing the trend of causes of death from 1900 until 1947, inclusive. (You are not responsible for producing the title and there are no other customizations done to the graphic beyond using the `datascience` library table methods with `causes_for_plotting`.)



Sample Solution:

```
(
    causes_for_plotting.where('Year', are.between(0, 1948))
    .plot('Year')
)
```

(b) Given the table `framingham_by_id`, write code that will store the value of the number of subjects in the study who had an average cholesterol level (`'AVE TOTCHOL'`) less than 200 in the variable `low_chol_level`.

Sample Solution:

```
low_chol_level = (
    framingham_by_id.where('AVE TOTCHOL',
                           are.below(200))
    .num_rows
)
low_chol_level
```

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Table Reference

Orders

Here is a preview of the table `orders`:

User	Restaurant	Total	Receiver	Rating	With Friends
w3ndyk1m	Sharetea	5.40	stephaniekeem	10	True
s_kw33	Riceful	10.24	haileyyebonjung	2	False
nikkyp	La Burrita	14.98	wfurtaco	7	True
sonyaki55	Poke Parlor	12.86	oskibear	8	False

. . . (46 rows omitted)

The table has 6 columns:

- **`User`**: (string) username of the user who purchased the order
- **`Restaurant`**: (string) restaurant name of the order
- **`Total`**: (float) total amount spent on the order, in dollars
- **`Receiver`**: (string) username of the user who received the order
- **`Rating`**: (int) how the user rated their order on a scale of 1-10 (10 being most satisfied)
- **`With Friends`**: (boolean) whether or not the user placed the order with friends

Heights

Here is a preview of the table `heights`:

- `height` is the **float** height of an individual, measured in inches
- `enrollee` is a **string** containing either "high school" or "college"

height	enrollee
68.5	college
62.1	high school
66.5	college

Skiers

Here is a preview of the table `skiers`:

Name	Sport	Height (in)	Downhill Time (s)	Favorite Resort
James	Ski	71	90.52	Vail
Eunice	Ski	66	93.64	Beaver Creek
Oscar	Snowboard	69	89.77	Heavenly
Rebecca	Snowboard	68	91.01	Palisades
Ciara	Ski	70	101.34	Park City

. . . (40 rows omitted)

Chocolates and Nutrition

The `chocolates` table contains 6 rows and 4 columns. The `'Color'` and `'Shape'` columns contains `str` values, the `'Inventory'` column contains `int` values, and the `'Unit Price ($)'` column contains `float` values.

Color	Shape	Inventory	Unit Price ($)
Dark	Round	4	1.3
Milk	Rectangular	6	1.2
White	Rectangular	12	2
Dark	Round	7	1.75
Milk	Rectangular	9	1.4
Milk	Round	2	1

The `nutrition` table contains 4 rows and 2 columns. The `'Type'` column contains `str` values and the `'Calories'` column contains `int` values.

Type	Calories
Dark	120
Milk	130
White	115
Ruby	120

Turtles

The `turtles` table contains 100 rows and 2 columns. The `'Temperature'` column contains `float` values and the `'Sex'` column contains `str` values.

Temperature	Sex
30.1525	Male
30.7116	Female
30.778	Female
29.2344	Male
31.961	Female

... (95 rows omitted)

Midterm Project

The `causes_for_plotting` table contains 124 rows and 5 columns. The `'Year'` column contains `int` values and the `'Cancer'`, `'Heart Disease'`, `'Stroke'`, and `'Unintentional Injuries'` columns contain `float` values.

Year	Cancer	Heart Disease	Stroke	Unintentional Injuries
1900	114.8	265.4	244.2	90.3
1901	118.1	272.6	243.6	109.3
1902	119.7	285.2	237.8	93.6
1903	125.2	304.5	244.6	106.9
1904	127.9	331.5	255.2	112.8

... (119 rows omitted)

The `framingham_by_id` table contains 4224 rows and 4 columns. The `'RANDID'`, `'AVE TOTCHOL'`, `'ANYCHD'`, and `'DEATH'` columns all contain `int` values..

RANDID	AVE TOTCHOL	ANYCHD	DEATH
2448	202	1	0
6238	249	0	0
9428	264	0	0
10552	228.5	0	1
11252	314	0	0

... (4219 rows omitted)